

Curcumin: a new paradigm and therapeutic opportunity for the treatment of osteoarthritis: curcumin for osteoarthritis management

The management of osteoarthritis represents a real challenge. This complex and multi-factorial disease evolves over decades and requires not only the alleviation of symptoms, i.e. pain and joint function but also the preservation of articular structure without side effects. Nutraceuticals are good candidates for the management of OA due to their safety profile and potential efficacy. However, they are not part of the treatment guidelines and published recommendations. Curcumin is the yellow pigment isolated from the rhizomes of *Curcuma longa*, commonly known as turmeric. Curcumin is a highly pleiotropic molecule with an excellent safety profile. Strong molecular evidence has been published for its potency to target multiple inflammatory diseases. However, naturally occurring curcumin cannot achieve its optimum therapeutic outcomes due to its low solubility and poor bioavailability. Nevertheless, curcumin presents great potential for treating OA and has been categorized as having preclinical evidence of efficacy. This review aimed at gathering most of the available information to document the potential efficacy of curcumin based on the results obtained in in vitro models of cartilage and osteoarthritis and in other diseases.